

TTB LabelVerify

Reviewer Testing Guide

Version 1.0 · June 2026 · Alcohol & Tobacco Tax and Trade Bureau

Live Application

<https://shatech4orward.github.io/ttb-labelverify>

Source Code

<https://github.com/shatech4orward/ttb-labelverify>

This document guides TTB/ATF compliance reviewers and technical evaluators through end-to-end testing of the TTB LabelVerify prototype. No software installation is required — the application runs entirely in a web browser.

1. What This App Does

TTB LabelVerify is a browser-based compliance verification tool designed to help TTB/ATF agents check whether alcohol product labels match their corresponding Certificate of Label Approval (COLA) applications. It works entirely offline inside the browser — no data is ever uploaded to a server.

The app uses optical character recognition (OCR) to read text directly from a label image, then compares that text against the fields in the COLA application. Think of it like a spell-checker for label compliance: it reads both documents and flags any mismatches.

Fields Checked

Field	What It Checks	Regulation
Brand Name	Label brand matches application; must be ALL CAPS	27 CFR 5.34
Class / Type	Exact designation (e.g. Kentucky Straight Bourbon Whiskey)	27 CFR 5.35
Alcohol By Volume	ABV within $\pm 0.15\%$ of application value	27 CFR 5.37
Net Contents	Volume within $\pm 3\%$ of application value	27 CFR 5.38
Name & Address	Bottler name and state match application	27 CFR 5.36
Country of Origin	Origin declaration present for imports	27 CFR 5.36(f)
Government Warning	Mandatory warning statement present and accurate	27 CFR 16.21

NOTE No internet connection is required after the page loads. All OCR processing happens locally in the browser.

2. How to Access the App

Open any modern web browser (Chrome, Firefox, Edge, or Safari) and navigate to:

<https://shatech4orward.github.io/ttb-labelverify>

No login, no account, and no software installation is required. The app loads directly in the browser. You will land on the Dashboard, which shows summary statistics and quick-access buttons to the three main features.

TIP For best results, use Google Chrome or Microsoft Edge. The app is mobile-friendly and also works on tablets.

Test Files Provided

Two zip archives are provided alongside this guide. Download both before testing:

File	Contents
TTB_Test_Labels.zip	8 label images (PNG) — 3 PASS, 2 WARNING, 3 FAIL
TTB_Sample_COLA_Applications.zip	8 COLA application PDFs matching the label images

3. Single Label Verification

Single label verification is the core feature. You upload one label image and one COLA PDF, and the app returns a full field-by-field compliance report within seconds.

Step-by-Step Instructions

Step 1 — Open Verify Label

Click the "Verify Label" button on the Dashboard, or click "Verify" in the left sidebar.

Step 2 — Select Beverage Type

Use the dropdown at the top to select the beverage category: Distilled Spirits, Wine, or Malt Beverage. For the sample files, select "Distilled Spirits."

Step 3 — Upload the Label Image

Click the label upload area (left panel) and select a PNG file from TTB_Test_Labels.zip. The image will preview immediately after upload.

Step 4 — Upload the COLA PDF

Click the COLA upload area (right panel) and select the matching PDF from TTB_Sample_COLA_Applications.zip. The app reads the application fields automatically.

Step 5 — Run Verification

Click the green "Verify Compliance" button. The OCR engine will scan the label image — this takes 5–15 seconds on first run (the OCR engine is loading). Subsequent runs are faster.

Step 6 — Review Results

A results panel appears below showing an overall score and a row for each of the 7 compliance fields. Each row shows PASS (green), WARNING (yellow), or FAIL (red) with an explanation.

Expected Results for Sample Labels

Label File	Product	Expected Result	Why
01_OLD_TOM_DISTILLERY_bourbon.png	Old Tom Distillery Bourbon	PASS	All fields match
02_SILVER_RIVER_VODKA.png	Silver River Vodka	PASS	All fields match
03_GLENCRAIG_SCOTCH_import.png	Glencraig Scotch (Import)	PASS	Import origin correctly declared
04_IRON_GATE_BOURBON_wrong_ABV.png	Iron Gate Bourbon	FAIL	ABV on label differs from application
05_BLUE_MESA_GIN_warning.png	Blue Mesa Gin	WARNING	Brand name not in required ALL CAPS

06_STONE_RIVER_RUM_brand.png	Stone River Rum	FAIL	Brand name on label doesn't match application
07_STONES_THROW_WHISKY.png	Stone's Throw Whiskey	WARNING	Capitalization inconsistency detected
08_COPPER_PEAK_TEQUILA.png	Copper Peak Tequila	FAIL	Government Warning statement missing

4. Testing with COLA PDF Applications

The app can parse COLA application PDFs directly — you do not need to manually type application data. The PDF reader extracts all compliance fields automatically.

Pairing Labels with COLA PDFs

Each label image in TTB_Test_Labels.zip has a matching COLA PDF in TTB_Sample_COLA_Applications.zip. The file names correspond:

Label Image	COLA Application PDF
01_OLD_TOM_DISTILLERY_bourbon.png	COLA_01_OLD_TOM_DISTILLERY.pdf
02_SILVER_RIVER_VODKA.png	COLA_02_SILVER_RIVER_VODKA.pdf
03_GLENCRAIG_SCOTCH_import.png	COLA_03_GLENCRAIG_SCOTCH.pdf
04_IRON_GATE_BOURBON_wrong_ABV.png	COLA_04_IRON_GATE_BOURBON.pdf
05_BLUE_MESA_GIN_warning.png	COLA_05_BLUE_MESA_GIN.pdf
06_STONE_RIVER_RUM_brand.png	COLA_06_STONE_RIVER_RUM.pdf
07_STONES_THROW_WHISKEY.png	COLA_07_STONES_THROW_WHISKEY.pdf
08_COPPER_PEAK_TEQUILA.png	COLA_08_COPPER_PEAK_TEQUILA.pdf

NOTE Always upload the label image first, then the COLA PDF. The app uses the COLA PDF to populate the application fields shown in the right panel.

5. Batch Upload Testing

The Batch Upload feature allows you to verify multiple labels at once — up to 200–300 labels in a single session. Results appear in a summary table when processing completes.

Step-by-Step Instructions

Step 1 — Open Batch Upload

Click "Batch Upload" in the left sidebar or on the Dashboard.

Step 2 — Select Beverage Type

Choose the beverage category from the dropdown. All labels in a batch must be the same type.

Step 3 — Upload Label Images

Click the label upload area and select all 8 PNG files from TTB_Test_Labels.zip at once. You can select multiple files by holding Ctrl (Windows) or Cmd (Mac) while clicking.

Step 4 — Upload COLA PDFs

Click the COLA upload area and select the matching PDFs. The app automatically matches label images to COLA PDFs by file name.

Step 5 — Start Batch Verification

Click "Start Batch Verification." A progress bar shows each label being processed. Processing all 8 labels takes approximately 60–90 seconds.

Step 6 — Review Batch Results

A results table appears showing one row per label with overall status, score, and any flagged fields. Click any row to see the full detail report for that label.

EXPECTED With all 8 sample files: 3 labels should show PASS (green), 2 should show WARNING (yellow), and 3 should show FAIL (red).

6. Testing with Your Own Files

After testing with the provided samples, you can upload real label images and COLA application PDFs to evaluate the tool against production data.

Supported File Formats

File Type	Accepted Formats	Notes
Label Image	PNG, JPG, JPEG, TIFF	Higher resolution = better OCR accuracy. Minimum 150 DPI recommended.
COLA Application	PDF	Standard TTB COLA PDF format. Text-based PDFs work best.

Tips for Best Results

- Use the highest resolution scan available — blurry or low-contrast images reduce OCR accuracy.
- Ensure the label image is flat and not photographed at an angle. Curved or distorted labels may cause OCR errors.
- The Government Warning text is small on most labels. A minimum 300 DPI scan is recommended for accurate detection.
- If a field shows an unexpected FAIL, scroll down in the results to see the raw OCR text captured — this helps identify whether the issue is OCR quality or a genuine compliance mismatch.
- Session data is not saved between browser sessions. Results are cleared when you close the tab.

7. Understanding Results

Result	Color	Meaning
PASS	Green	The label field matches the COLA application within acceptable tolerances.
WARNING	Yellow	A minor deviation was detected — e.g., capitalization issue or near-threshold value. Human review recommended.
FAIL	Red	A significant discrepancy was detected between the label and the application. The label does not comply.

Score

Each verification produces an overall compliance score from 0–100%. The score reflects the proportion of fields that passed. A WARNING counts as a partial pass. A FAIL reduces the score proportionally. A score of 100% means all fields passed.

Agent Override

On the detail page for any verification, agents can apply a manual override to change the overall result — for example, if OCR quality was poor and a human review confirms the label is actually compliant. Overrides are logged in the Review Queue.

8. Technical Notes (For Evaluators)

Architecture Overview

Component	Technology	Purpose
Frontend Framework	React 18 + TypeScript	UI components and application logic
Build Tool	Vite 7	Bundles and optimizes the client app
OCR Engine	Tesseract.js v7 (WASM)	Reads text from label images — runs entirely in browser, no server calls
PDF Parser	pdfjs-dist v5.6	Extracts text from COLA application PDFs client-side
Compliance Engine	Custom TypeScript (labelVerifier.ts)	Fuzzy matching + numeric tolerance checks against TTB regs
Hosting	GitHub Pages	Free static hosting — no backend server required
Styling	Tailwind CSS + shadcn/ui	Responsive layout, accessible components

Privacy & Data Handling

All processing occurs locally in the user's browser. Label images and COLA PDFs are never transmitted to any external server. No user accounts, cookies, or tracking are used. Session data (job history) is stored in browser memory and cleared on page close.

Matching Thresholds

Field	PASS Threshold	WARNING Threshold
Brand Name	≥97% similarity, ≤2 char difference	≥88% similarity, ≤2 char difference
Class / Type	≥97% similarity, exact length match	≥90% similarity, exact length match
ABV (Spirits)	Within ±0.15%	N/A (FAIL if outside)
Net Contents	Within ±3%	N/A (FAIL if outside)
Name & Address	Full ≥88%, Company ≥92%, State ≥90%	Any component below PASS threshold
Govt Warning	≥92% similarity	≥78% similarity

Known Limitations

- OCR accuracy depends on image quality. Scanned labels at low DPI may produce false failures.
- The Government Warning check requires the warning to begin on a separate line from surrounding text.

- Handwritten or stylized typeface labels may score lower due to OCR character recognition limits.
- This is a prototype — results should be treated as decision-support tools, not final determinations.

Source Code

The full source code is publicly available at: <https://github.com/shatech4orward/ttb-labelverify>